

Smart Medicine Reminder with a Mobile Application

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Abstract -- Medication non-adherence is a major healthcare concern, especially among elderly and chronic disease patients, often resulting in poor treatment outcomes. This paper presents an IoT-based Smart Medicine Reminder System designed to improve medication adherence through automated reminders and monitoring. The system uses an Arduino/ESP32 microcontroller, real-time clock, buzzer, LEDs, and a smartphone application to provide timely medication alerts and maintain dosage records. Inspired by the work of Lee et al. (2019), the proposed solution supports multiple medications, adherence tracking, snooze functionality, and caregiver notifications for missed doses. Medication intake data are securely stored in a cloud database, enabling remote monitoring and analysis. The system architecture, hardware and software requirements, implementation approach, and testing procedures are discussed. By combining low-cost IoT technology with user-friendly features, the proposed system aims to enhance medication compliance, support independent living, and improve overall healthcare outcomes.

Keywords : Healthcare Automation, Scheduling, Medication Adherence, Monitoring, Android Application, Chronic Disease Management.

I. INTRODUCTION

Patients who are elderly or those on complex regimens often forget or mix up medications. Such errors can exacerbate conditions or cause hospitalizations. Manual methods require effort. Advances in IoT and mobile apps now enable smart reminders. We design an integrated system where an Arduino/ESP32-based reminder alerts the user at each dose time, and a mobile app manages schedules and notifications. The system is intended as a college mini-project prototype but follows rigorous engineering and research standards. Technology has become a part of our daily life and as a result the living as well as individual demands have increased. People who are taking medications on daily basis represent a significant proportion of the community all over the world. With such a large amount of prescribed medicine, the probability of forgetting to take medications at a particular time or according to the amount prescribed is high. The American Society of Health System Pharmacists found that about one third of older adults are prescribed to take eight or more medications each day. With such a large amount of prescriptions, the consequences of forgetting to take

medications at a particular time or, worse, accidentally taking the same medication twice are significantly high. The British Pharmacological Society noted that greater than 80 percent of elderly are hospitalized due to harmful drug reactions caused by dosage errors. While prescription bottles have strict instructions covering their surfaces, all of these reminders are futile in patients struggling with memory pathologies, such as Alzheimer. Based on a report from the United Nations in Ageing and Development 2009, the average annual growth rates of population aged 60 or over are very high. Many of old people live alone, some of them are suffering from disability, making it harder to take care of themselves. Any delay or ignorance such as forgetting to take medicines or even taking medicines at the wrong time may cause or raise potential health issues. It is also difficult for the nurses to know recent information regarding the number of pills remaining and the need to refill some medications. Therefore, there is an increasingly desperate need for something that can improve life quality for elderly people. For example, smart devices that are invented to detect vital signs of a person, or to send emergency messages to a healthcare professional when urgent help is needed, or to diagnose and determine the causes of certain diseases after ingesting into patients' bodies.

II. LITERATURE SURVEY

Initial research in this domain focused on basic reminder systems. The Future Computing and Informatics Journal (2018) discussed early smart pill reminder solutions that relied on alarms and notifications to alert patients. While these systems were simple and cost-effective, they depended heavily on patient response and lacked tracking or monitoring capabilities.

[1] Chowdhury et al. (2020) proposed a medication reminder system tailored for elderly and visually impaired individuals. The system ensured timely alerts through audio and visual cues, improving adherence to some extent. However, it was designed for single-user scenarios and did not support caregiver monitoring or multi-patient environments.

[2] Lee et al. (2019) developed an IoT-based medication management system that used sensors and notifications to remind patients and log their responses. This system improved adherence tracking and introduced basic monitoring features. However, it lacked scalability and centralized control, limiting its usability in healthcare institutions.

[3] Anwar et al. (2020) introduced a smart medication reminder system with compliance tracking, which recorded whether patients followed their schedules. Although effective in identifying missed doses, the system was limited to individual use and lacked support for multiple patients or caretaker dashboards.

[4] Research by Vasisht et al. (2020) explored wearable healthcare technologies that track medication adherence through activity monitoring and reminders. These systems provided continuous monitoring and were particularly useful for remote healthcare. However, they relied heavily on user interaction and did not provide centralized management features.

[5] Pang et al. (2023) proposed a mobile-based IoT medication reminder system focusing on portability and user convenience. The system allowed patients to receive reminders on their smartphones and track their medication schedules. Despite its advantages, it did not include caregiver monitoring or shared system capabilities.

[5] Siddiqha et al. (2024) developed an IoT-based pill reminder and monitoring system capable of sending alerts and maintaining basic logs. While practical for individual users, it lacked cloud-based storage and real-time access for caregivers, reducing its effectiveness in institutional use.

[6] Patel (2023) analyzed multiple IoT-based reminder systems and concluded that most solutions focus on individual adherence improvement, with limited exploration of centralized cloud-based monitoring and multi-user systems.

[7] Abhijith et al. (2022) proposed a low-cost Arduino-based medication reminder system aimed at affordability and accessibility. While effective for basic reminder functionality, it lacked advanced features such as remote monitoring, cloud integration, and patient identification.

The system discussed in SN Computer Science demonstrated improved adherence using IoT-enabled reminders and tracking. However, like many others, it remained limited to a single-user setup and did not support scalable deployment.

From the above literature survey, it is observed that most existing systems focus on automation and alert mechanisms using IoT and microcontrollers. However, many systems lack feedback verification, reliability during power or network failure, and user-friendly design.

The proposed system overcomes these limitations by integrating:

- ESP32 for efficient processing and connectivity
- RTC module for accurate timekeeping
- Sensor-based feedback for reliability
- Alert system using buzzer and LEDs

Thus, the proposed design provides a more reliable, cost-effective, and user-friendly smart pill dispensing solution.

III. METHODOLOGY

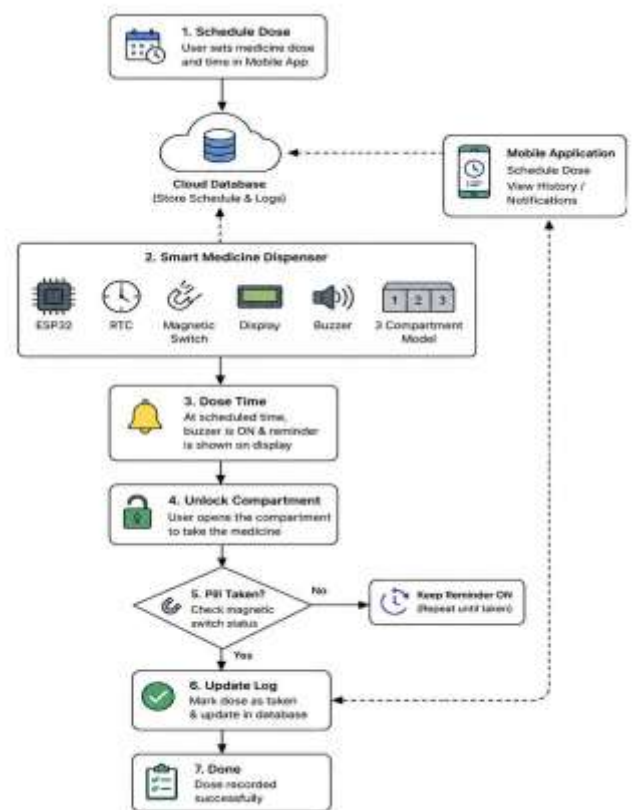


Fig1: Flow chart of design and implementation

The implementation of the Smart Medicine Reminder System is based on an embedded IoT architecture using the ESP32 microcontroller. The system is designed to automate the process of medicine consumption, ensure timely alerts, and verify medication intake through sensors. The methodology involves hardware integration, software control, and real-time monitoring.

1. System Initialization

At the start, the ESP32 microcontroller initializes all connected peripherals including:

- RTC module (DS3231)

- LCD display (16×2 I2C)
- Servo motors
- IR sensors / limit switches
- Buzzer and LEDs

The system checks for proper connectivity of all components. The RTC module is synchronized with the current date and time, either manually or via network time (NTP). The predefined medication schedule is stored in the ESP32 memory.

2. Time Monitoring and Scheduling

The RTC continuously provides real-time data to the ESP32 through I2C communication (SDA – GPIO 21, SCL – GPIO 22). The microcontroller compares the current time with the stored medication schedule.

If the current time matches the scheduled time → trigger dispensing process

If no match → system remains in monitoring mode

This continuous time-checking ensures precise scheduling without delays.

3. Alert Generation System

Buzzer produces an audible alert

LED indicators glow to indicate active dispensing

LCD display shows messages such as:

Time to take medicine

Medicine dispensed

This multi-modal alert system ensures that the user is notified effectively, even in noisy or low-visibility environments.

4. IoT Monitoring and Data Logging

Using the ESP32's built-in Wi-Fi:

The system sends data to a cloud server or mobile application
Information logged includes:

Status (Taken / Missed)

System alerts Caregivers or users can monitor medication adherence remotely. Notifications can also be sent via mobile apps or web dashboards.

5. Power Management

The system operates using a dual power supply:

5V supply for servo motors and LCD

3.3V supply for ESP32

A regulated power supply ensures stable operation. In case of power failure:

RTC continues operation using battery backup

System resumes normal function after power restoration

6. Continuous Operation Loop

The system runs in a continuous loop:

Read RTC time

Compare with schedule

Activate alerts

Verify using sensors

Log data and update status

Return to monitoring state

This loop ensures uninterrupted and real-time operation of the system.

7. System Reliability Features

To enhance performance and reliability:

Debouncing techniques are used for sensors

Servo calibration ensures accurate positioning

Error handling mechanisms detect hardware failures

Watchdog timers prevent system crashes

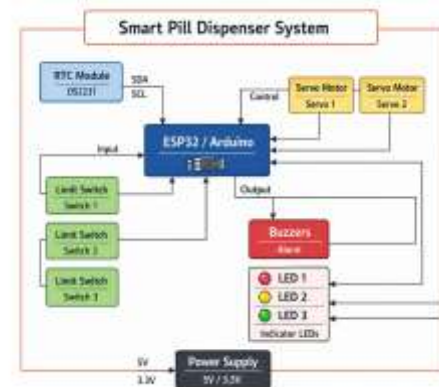
8. Overall Workflow Summary

The complete system workflow can be summarized

Initialization → Time Monitoring → Alert → Verification →

Logging → Repeat

This structured methodology ensures efficient, accurate, and reliable medication management



BLOCK DIAGRAM.

The Smart Medicine Reminder System consists of an ESP32 microcontroller as the central control unit, which coordinates all system operations. A DS3231 RTC module provides accurate real-time data for scheduling medication. The ESP32 communicates with the RTC and the 16×2 LCD display using the I2C protocol to display time and system status. Based on the scheduled time, the ESP32 controls alert the users. IR sensors or limit switches are used to detect user interaction, ensuring proper operation. An alert system comprising a buzzer and LEDs notifies the user when it is time to take medication. The entire system is powered by a regulated 5V/3.3V power supply, and with built-in Wi-Fi, the ESP32 enables IoT-based monitoring and data logging.

EXPERIMENTAL SETUP



The experimental setup of the Smart Medicine Reminder System is designed to validate the functionality, accuracy, and reliability of the proposed model. The system is built using an ESP32 microcontroller mounted on a development board, which acts as the central control unit. A DS3231 RTC module is connected via I2C communication to provide precise timekeeping for medication scheduling. A 16×2 LCD display (I2C-based) is interfaced to display real-time information such as current time, system status, and alerts.

The pill reminding mechanism is implemented to remind pills dosage at scheduled times. IR sensors or limit switches are placed inside the compartment to detect pill status.

An alert system, consisting of a buzzer and LED indicators, is integrated to notify users at scheduled time. The entire setup is powered using a regulated power supply providing both 3.3V (for ESP32) and 5V (for motors and display). The ESP32 is programmed using Arduino IDE, where the scheduling logic, sensor feedback, and control operations are implemented.

For testing, multiple time schedules are configured, and the system is observed for correct alerts on scheduled timings. Data such as response time, dispensing accuracy, and system reliability are recorded to evaluate performance. This setup ensures a practical demonstration of automated medication management with real-time monitoring capabilities



Fig. When the pill compartment is opened to take the pill

ADVANTAGES

i. Remind the Patient to Take the Right Medication and Right Dose at the Right Time, Every Day: People of all ages forget to take medication easily, especially those with busy lives and varying schedules. A Pill Reminder will remember for you and remind you, so you will never forget to take your medications again!

ii. Prevent Medication Errors: It can be easy for you or a loved one has to take the wrong medication or skip doses. With an Automatic Pill Reminder, the patient does not have to worry about if they are taking the correct meds since they are pre-sorted.

iii. Make the Patient and the Caregiver Feel Better: Not only will taking the medication more consistently make the patient feel better, but the caregiver will have peace of mind!

FUTURE ENHANCEMENT

From the time the pills were introduced in the medication industry the requirement of a device that can come as a substitute to the human being for delivery of medicine. No pill dispenser to today could replace could determine the person who to give a pill to, when to give a pill and the no. of pills to be given. There are many other problems that plague the home pill user, such as someone stealing pills, forgetting to take them or having the pills available to take too many. AMDR mechanism such that it works on the principle of MECHATRONICS that is it combines the mechanics as well electronics to dispense medicine. Therefore, chances of error are very less. Following are some of the sector that can be made better by AMDR: **i. Hospitals:** –The automatic pill dispenser can ease up the pill dispensing operation of the large as well as small health care centre. With the help AMDR, the medical staff, as well as a doctor, can reduce the amount of time as well frequent check-up of every patient. This will help the doctor to give time to those patients which are in critical condition or are in greater need for attention from the medical staff. **ii. Home:** – Vitamins and another essential dietary supplement that is required every day before and after a meal can become a great area of interest for the use of an automatic pill dispenser. An individual keychain key can initiate the pill dispensing for the user's personal diet regime and a buzzer can remind them when the supplements need to be taken. **iii. Elderly:** –The automatic pill dispenser is an excellent way for the elderly to be reminded when to take their pills and the current access key can be worn around the neck or on a bracelet.

CONCLUSION

The Smart Medicine Reminder System successfully demonstrates an efficient and reliable solution for automated medication management using embedded systems and IoT technology. By integrating the ESP32 microcontroller with a real-time clock, sensors, and alert systems, the proposed system ensures timely and accurate delivery of medicines. The use of feedback mechanisms such as IR sensors enhances reliability by confirming whether the medicine has been taken.

The system reduces dependency on human memory, minimizes medication errors, and improves adherence, especially for elderly and chronically ill patients. Additionally, the integration of Wi-Fi enables real-time monitoring and data logging, making the system more effective for caregivers and healthcare providers. Overall, the project proves to be a cost-effective, user-friendly, and scalable solution for modern healthcare applications.

FUTURE SCOPE

In the future, the Smart Medicine Reminder System can be enhanced by integrating advanced technologies to improve functionality and usability. A dedicated mobile application can be developed to allow users and caregivers to remotely monitor schedules, receive notifications, and control the system. Incorporating artificial intelligence (AI) can enable smart dosage recommendations and predictive analysis based on patient history.

The system can also be upgraded with voice assistant integration to assist elderly users in interacting with the device more easily. Adding biometric authentication such as fingerprint sensors can improve security and prevent unauthorized access to medicines. Furthermore, integration with cloud-based healthcare systems and hospital databases can enable real-time health tracking and medical record management.

Additional improvements may include battery backup systems for uninterrupted operation, compact design for portability, and enhanced sensor systems for more accurate monitoring. These advancements can make the system more intelligent, secure, and widely applicable in real-world healthcare environments.

REFERENCES

- [1] S. C. Lee, Y. J. Park, and J. W. Lee, "IoT-Based Smart Medication Management System for Elderly Patients," *Sensors*, vol. 19, no. 17, p. 3657, 2019.
- [2] K. P. R. Chowdhury, M. A. Rahman, and S. Hossain, "Design and Implementation of an Automated Pill Dispenser for the Elderly and Visually Impaired," *IEEE Access*, vol. 8, pp. 151567–151579, 2020.
- [3] M. R. Anwar, T. Rahman, and M. S. Islam, "Development of a Smart Medicine Dispenser with Patient Compliance Monitoring," *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 11, no. 6, 2020.
- [4] D. K. Vasisht, A. Shukla, and N. Gupta, "Smart Healthcare Devices for Efficient Medication Adherence Monitoring," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 7, pp. 2029–2039, 2020.
- [5] W. L. Pang, K. C. Ong, and C. Y. Chua, "Smart IoT Mobile Medication Dispenser (SMMD)," *International Journal of Intelligent Systems and Applications in Engineering (IJISAE)*, 2023.
- [6] A. Siddiqha, R. Karthikeyan, and N. Kumar, "IoT-Based Pill Reminder and Monitoring System," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, 2024.

[7] P. Patel, "IoT-Based Smart Medication Dispenser Systems for Elderly Care," *Journal of Machine Learning and Health Data Science (JMLHDS)*, vol. 3, no. 1, 2023.

[8] Abhijith M., S. R. Nair, and P. Varma, "IoT-Based Smart Medicine Dispenser: A Technological Solution for Medication," *Journal of Neonatal Surgery*, 2022.

[9] "A Novel IoT-Based Medicine Consumption System for Elders," *SN Computer Science*, Springer, 2022.

[10] "Smart Drugs: Improving Healthcare Using Smart Pill Box for Medicine Reminder and Monitoring System," *Future Computing and Informatics Journal*, vol. 3, issue 2, 2018